

Plant Disease Classification Using CNN

Technical Summary Report

Summary

This project develops a deep learning-based computer vision solution for classification of plant diseases by using a Convolutional Neural Network (CNN). The model achieves strong classification performance on agricultural leaf disease detection, trained on the PlantVillage dataset containing 20,639 images across 15 disease classes.

Objective

The objective is to build and train a CNN model to classify plant leaf images into healthy or diseased categories across 15 different disease conditions for early disease detection and intervention.

Dataset Characteristics

Metric	Details
Dataset Source	PlantVillage
Total Images	20,639 images
Number of Classes	15 disease classes (including healthy plants)
Average Images/Class	about 1,375 images per class
Class Balance	Moderate imbalance (150-3,000 images per class)
Data Quality Issues	1 unreadable file identified; auto-filtered during preprocessing

Model Architecture

Framework & Libraries

TensorFlow 2.x & Keras with scikit-learn for model evaluation and metrics

Network Architecture

Sequential CNN with the following layers:

- Input Layer: 224×224×3 RGB images
- Convolutional Block 1: 32 filters (3×3 kernel), ReLU activation, BatchNormalization, MaxPooling (2×2)
- Convolutional Block 2: 64 filters (3×3 kernel), ReLU activation, BatchNormalization, MaxPooling (2×2)
- Convolutional Block 3: 128 filters (3×3 kernel), ReLU activation, BatchNormalization, MaxPooling (2×2)
- Fully Connected Layers: Dense layers with dropout for regularization
- Output Layer: 15 units (softmax activation) for multi-class classification

Training Configuration

- Optimizer: Adam (learning rate = 0.001)
- Loss Function: Sparse Categorical Crossentropy
- Metrics: Accuracy
- Epochs: 30 with Early Stopping callback
- Regularization: Batch Normalization to stabilize training

Results & Performance Evaluation

Test Set Performance

- Confusion Matrix: Generated for detailed per-class performance analysis
- Classification Report: Precision, recall, and F1-scores computed for all 15 classes

Inference Example

- Top Prediction: Tomato Target Spot (81.22% confidence)
- Runner-up Predictions: Tomato Early Blight (6.94%), Tomato Spider Mites (4.68%)
- Result: Model correctly classified unseen test image with high confidence

Model Deployment

- Trained model saved as: `plant_disease_cnn.keras`
- Model can be loaded for inference on new images using `Keras load_model()`
- Prediction function supports top-3 predictions with confidence scores

Key Findings & Observations

- Visual Pattern Recognition: Healthy leaves are uniform green; diseased leaves show distinct lesions, discoloration, or irregular patterns
- Disease Diversity: Each disease presents unique visual symptoms (e.g., dark brown lesions for Early Blight, mold patterns for Leaf Mold)
- Dataset Balance: Moderate class imbalance successfully handled; Tomato Yellow Leaf Curl Virus has 3,000+ images vs. Healthy Potato with ~150
- Data Quality: Robust preprocessing pipeline handles corrupted files without manual intervention

Recommendations & Future Work

- Model Optimization: Experiment with transfer learning (ResNet, VGG, EfficientNet) for improved generalization
- Class Imbalance: Implement class weighting or augmentation techniques to address dataset imbalance
- Robustness Testing: Evaluate model on real-world images with varying lighting, angles, and environmental conditions
- Production Deployment: Convert to mobile-optimized format (TFLite) for field deployment on agricultural devices
- Explainability: Implement Grad-CAM or attention visualization for model interpretability

Conclusion

This CNN-based approach demonstrates the feasibility of automating the classification of plant disease with high confidence. The model successfully learns distinctive visual features across 15 disease classes and achieves strong predictive performance. The modular architecture and saved model weights enable easy deployment for practical agricultural applications such as precision farming, early disease detection, and intervention planning.

The github link to this project is : https://github.com/ToluDan/Project_Disease_Classifier